

1. End-to-End Execution Sequence

1. Purchase order telemetry and telemetry logs stream via Kafka into PostgreSQL.
2. Feature Store calculates lead-time variance, defect rates, and price volatility.
3. XGBoost model evaluates features and updates dynamic risk score (0-100).
4. Risk scores ≥ 75 trigger mobile push notifications and populate the Next.js workspace queue.
5. User adjusts sliders in What-If simulator; SciPy solver outputs re-allocation schedule.
6. LangChain pipeline synthesizes audit logs and outputs executive procurement report to MongoDB.